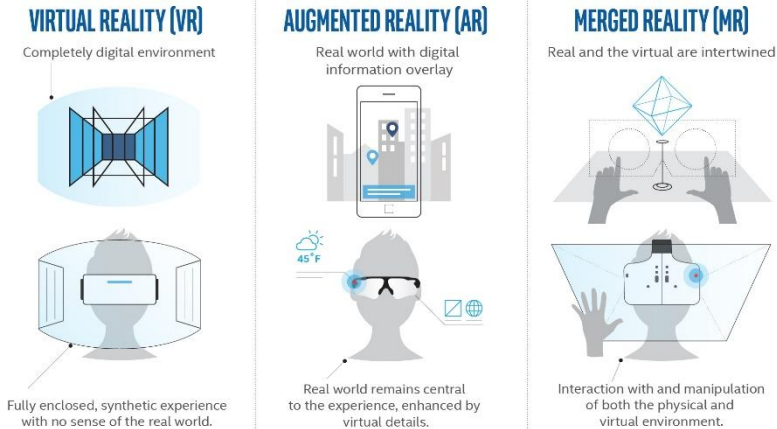


AR Lecture 4



AR vs VR vs MR

**Presented
by Prof. V. Karthikeyan**

Why Comparison Matters?

- Rapid growth of immersive technologies.
- Incorrect technology choice leads to high cost and low usability.
- Need to understand **where AR fits vs VR vs MR.**

AR vs VR vs MR – Conceptual Overview

- **Definitions Recap:**
- **AR:** Enhances real world with digital content.
- **VR:** Replaces real world with virtual environment.
- **MR:** Blends real and virtual worlds with interaction.
- **Core Difference:**
- Level of **immersion**
- Level of **real-world awareness**

Detailed Comparison: AR vs VR vs MR

Feature	AR	VR	MR
Real World Visibility	Yes	No	Yes
Immersion Level	Low–Medium	High	Medium–High
User Isolation	No	Yes	Partial
Interaction	Limited	Full	Advanced
Hardware	Mobile, Glasses	Headsets	HoloLens
Cost	Low	Medium–High	High
Use Environment	Real world	Virtual world	Hybrid

Selection of Technology: AR or VR

1. Application Objective



Visualisation



Simulation

2. User Environment



Controlled Environment



Real World operation

3. Level of Immersion Required



Low



High

4. Hardware Availability



Dedicated Headset & Haptics



Mobile Devices

5. Cost and Scalability



High end experience



Low budget and scalable

Case study

Case 1: Education



VR



AR

Case 2: Industry



VR



AR

Recap

- **Key Takeaways:**
- AR adds, VR replaces, MR blends
- Selection depends on use case, cost, and immersion
- No single technology fits all problems